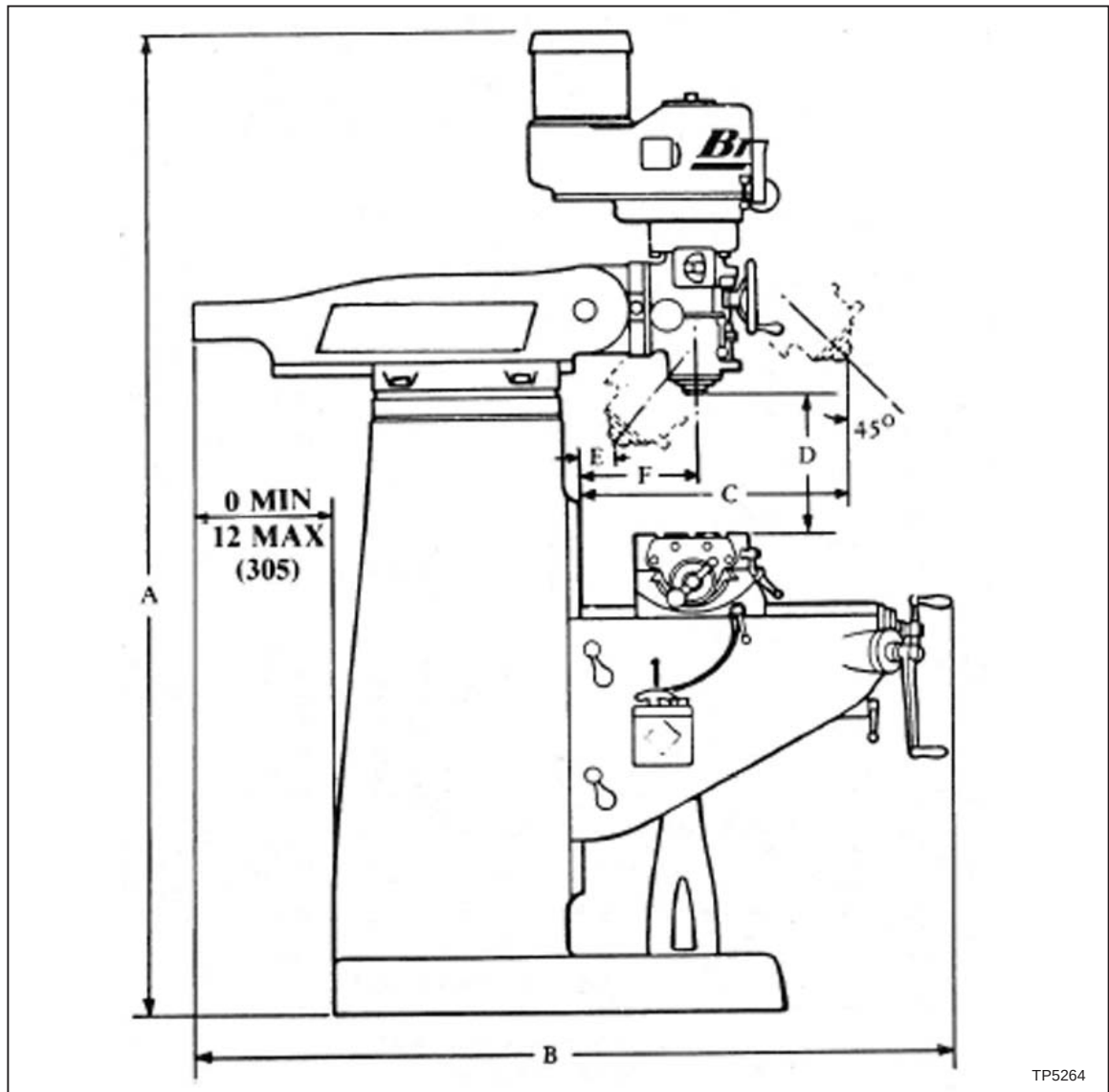


CHAPTER 5 - SPECIFICATIONS

MACHINE

PRINCIPLE DIMENSIONS



TP5264

Figure 5.1 - Machine Principle Dimensions

TABLE TRAVEL

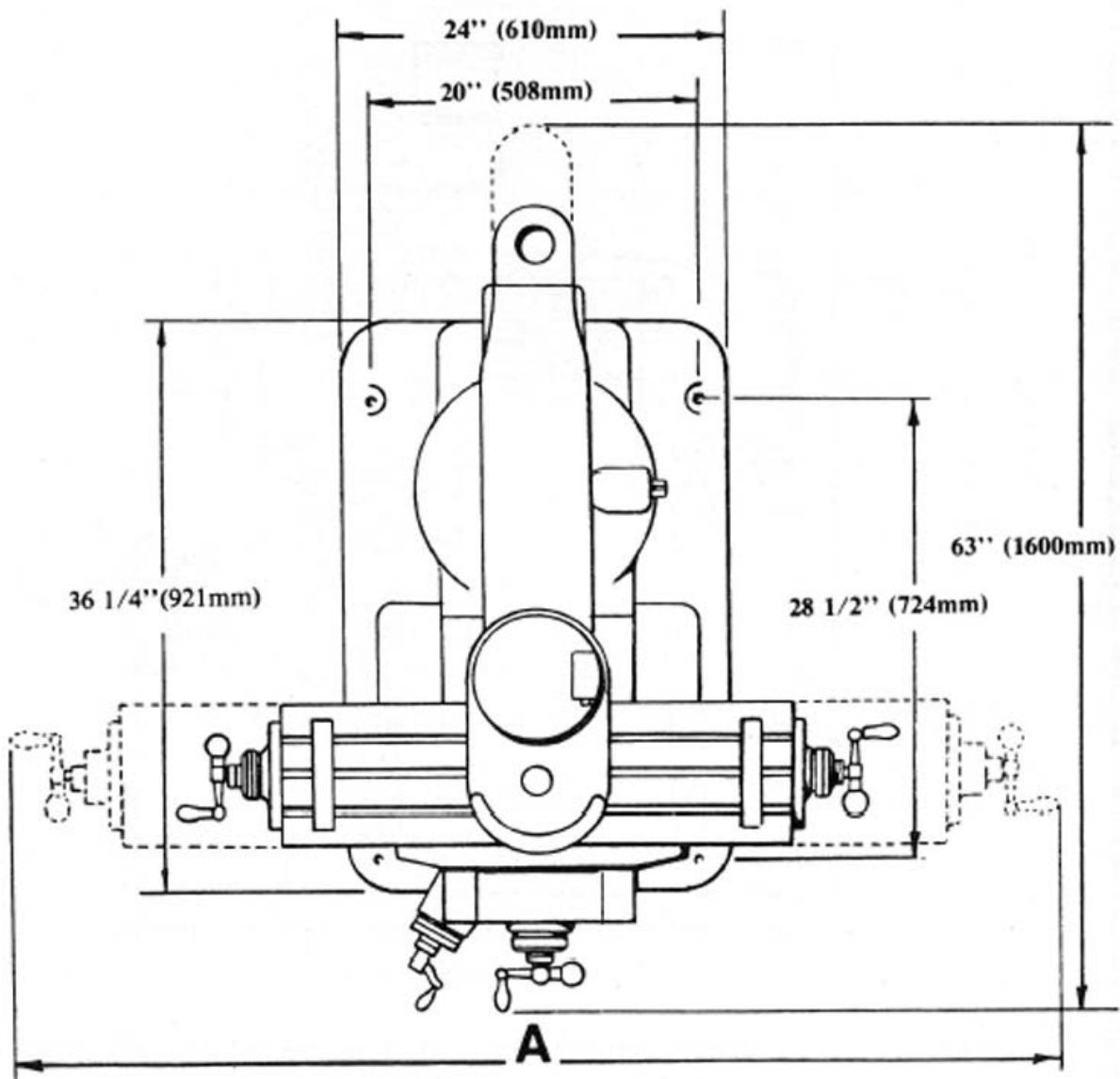
30" (762mm)
36" (914mm)

TABLE LENGTH

42" (1067mm)
48" (1219mm)

	A	B	C	D	E	F
MIN	82" (2083)	51" (1295)	8.750" (222)	2.500" (64)	0	6.750" (171)
MAX	82" (2083)	63" (1600)	20.750" (527)	18.500" (470)	12" (305)	18.750" (476)

Note: Metric specifications in parenthesis



TP5265

Figure 5.2 - Machine Specifications
Principle Dimensions

'A' DIMENSION

TABLE SIZE

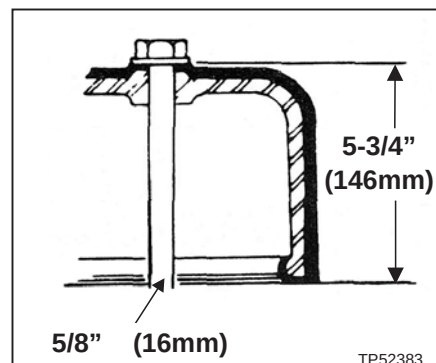
42" (1067mm)
48" (1219mm)

PLAIN MACHINE

87" (2210mm)
93" (2362mm)

POWER FEED

91" (2311mm)
97" (2464mm)



TP52383

Figure 5.3 - Machine
Dimension - Hold-Down Bolt

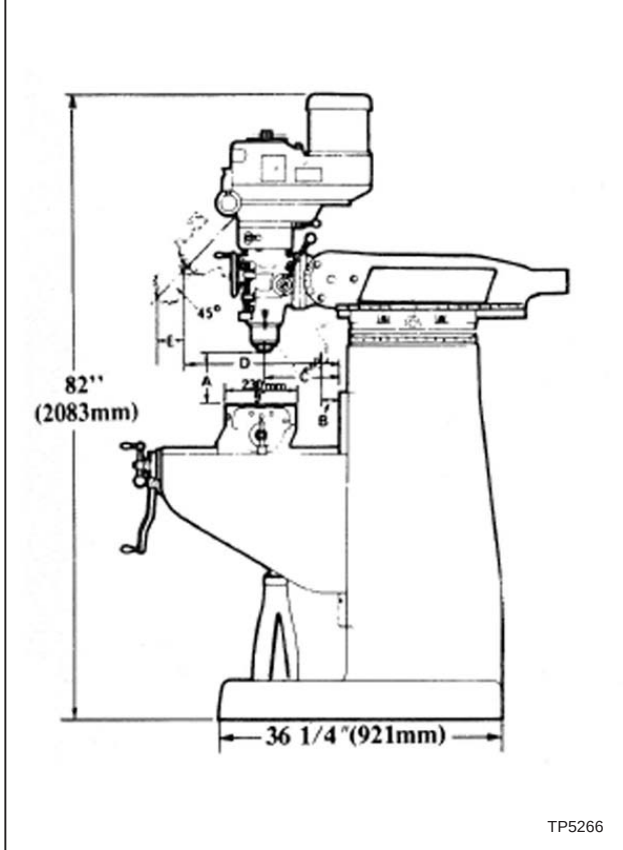


Figure 5.4 - Machine Specifications
Side View

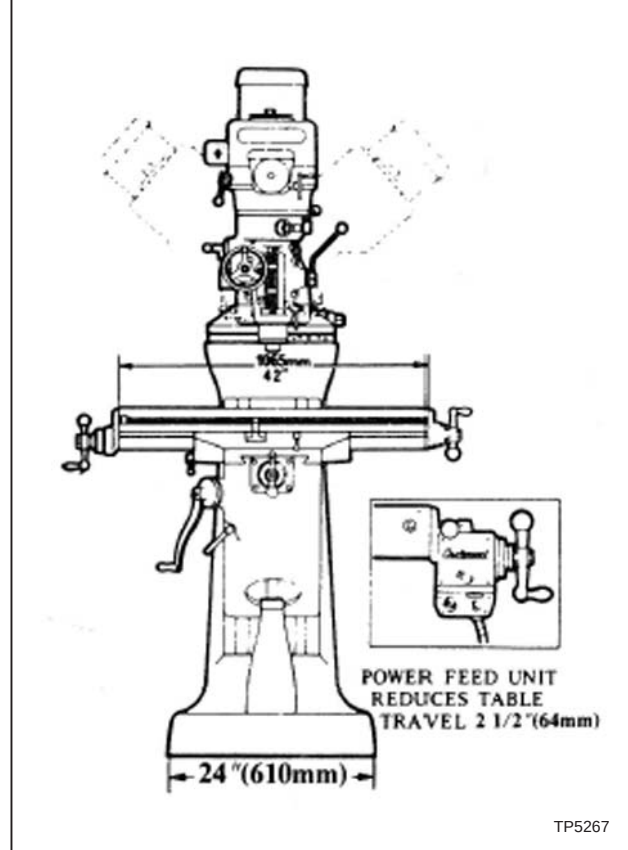


Figure 5.5 - Machine Specifications
Front View

Table Size	42" x 9"	1067mm x 229mm
Table Travel Hand Only	30"	762mm
Table Travel with Power Feed	27.500"	699mm
Saddle Travel	12"	305mm
Vertical Travel	16"	406mm
Number of Tee Slots	3	3
Width and Centers	.625" at 2.500"	15.9mm at 63.5mm
Ram Movement	12"	305mm
Spindle Speeds-RPM (Infinitely Variable)	60 – 4200	
Spindle Center to Column Face	6.750" – 18.750"	171mm – 476mm
Spindle to Table	2.500" – 18.500"	64mm – 470mm
Quill Feeds per Rev of Spindle	.0015" - .003" - .006"	.04mm, .08mm, .15mm
Shipping Particulars, Space Occupied	115 cu. Ft.	3.26 cu. M
Net Weight (Basic Machine)	1988 lb.	900 kg
Gross Weight (Basic Machine)	2604 lb.	1180 kg

Machines can also be fitted with a 48" x 9" (1219m x 228mm) table with a travel of 36" (914mm). If an X-Axis power feed unit is selected, the table travel is reduced by 2.500" (64mm).

MACHINE SPECIFICATIONS

RANGE

Table Travel (X-Axis)	30" (762mm) (42" Table)	36" (914mm) (48" Table)
Saddle Travel (Y-Axis)	12" (305mm)	
Quill Travel	5" (127mm)	
Knee Travel (Z-Axis Manual)	16" (406mm)	
Ram Travel	12" (305mm)	
Throat Distance		
(Minimum)	6.75" (171mm)	
(Maximum)	18.75" (476mm)	
Table to Spindle Nose Gage Line		
(Minimum)	2.25" (57mm)	
Maximum Weight of Workpiece	750 lb. (340kg)	

TABLE

Overall Sizes	9" x 42" (229mm x 1067mm)	9" x 48" (229mm x 1219mm)
T-Slots	3 on 2.5" (64mm) Centers	
T-Slot Size	.625" (16mm)	
Height Above Floor (Max.)	47.75" (1213mm)	

MILLING

STD POWER FEED

HIGH TORQUE
POWER FEED

Feed Rate*	(X) .750-35 IPM (19-889mm/min)	(X) .375-15 IPM (9.5-381mm/min)
------------	-----------------------------------	------------------------------------

SPACE AND WEIGHT

Floor Area	7' x 10' (2.1m x 3.1m)
Height	82" (2083mm)
Net Weight	1988 lb. (900kg)
Shipping Weight	2180 lb. (989kg)

POWER

Electric Supply – 60 Hz, 3-Phase	208/230/460/575v
----------------------------------	------------------

COLOR

Standard	Standard Machine Gray
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*Power optional – power feed on X-table or Y cross feed is optional equipment.

MILLING SPECIFICATIONS

*No longer available/part orders only

MODEL	M-HEAD	J-HEAD	J-HEAD HI-SPEED	2J-HEAD PRE-1977	2J-HEAD POST-1977
Power	.5 HP	1.0 HP	1.5 HP	1.5 HP	2.0 HP
Motor RPM	1200 RPM	1800 RPM	3600 RPM	1800 RPM	1800 RPM
Speed Ranges –RPM	6 Steps	8 Steps	8 Steps	Stepless	Stepless
Low	275 – 4550	80 – 325	160 – 660	60 – 500	60 – 500
High		660 – 2720	1320 – 5440	500 – 4200	500 – 420
Quill Travel	3.5" (88.9mm)	5.0" (127mm)	5.0" (127mm)	5.0" (127mm)	5.0" (127mm)
Quill Diameter	2.562" (65mm)	3.375" (86mm)	3.375" (86mm)	3.375" (86mm)	3.375" (86mm)
Spindle Taper	#2 Morse #7 B and S B-3	R-8 #30 QC	R-8 #30 QC	R-8 #30 QC	R-8 #30 QC
Spindle Diameter	1.437" (36.5mm)	1.865" (48mm)	1.875" (48mm)	1.875" (48mm)	1.875" (48mm)
Spindle Feed Rate	Manual	.0015/Rev (.038mm) .003/Rev (.076mm) .006/Rev (.152mm)	.0015/Rev (.038mm) .003/Rev (.076mm) .006/Rev (.152mm)	.0015/Rev (.038mm) .003/Rev (.076mm) .006/Rev (.152mm)	.0015/Rev (.038mm) .003/Rev (.076mm) .006/Rev (.152mm)
Drilling Capacity Manual	.50" Dia (12.7mm)	.75" Dia (19mm)	.75" Dia (19mm)	.75" Dia (19mm)	.87" Dia (22mm)
Power		.37" Dia (9.4mm)	.37" Dia (9.4mm)	.37" Dia (9.4mm)	.37" Dia (9.4mm)
Boring Capacity	1.50" Dia (38mm)	6.0" Dia (152.4mm)	6.0" Dia (152.4mm)	6.0" Dia (152.4mm)	6.0" Dia (152.4mm)
Milling Capacity	1.0" ³ /Min (16 cc/Min)	1.5" ³ /Min (24 cc/Min)	1.5" ³ /Min (24 cc/Min)	2.0" ³ /Min (32 cc/Min)	2.0" ³ /Min (32 cc/Min)
Spindle to Column Minimum	7.5" (190.5mm)	6.0" (152mm)	6.0" (152mm)	6.0" (152mm)	6.0" (152mm)
Maximum	19.0" (483mm)	23.0" (584mm)	23.0" (584mm)	23.0" (584mm)	23.0" (584mm)

RECOMMENDATIONS:

Use 2, 3 or 4 end mills. Eight-flute end mills are usually not satisfactory for general milling. When using shell mills, face mills or any other tooling, proper machining practice should be observed.

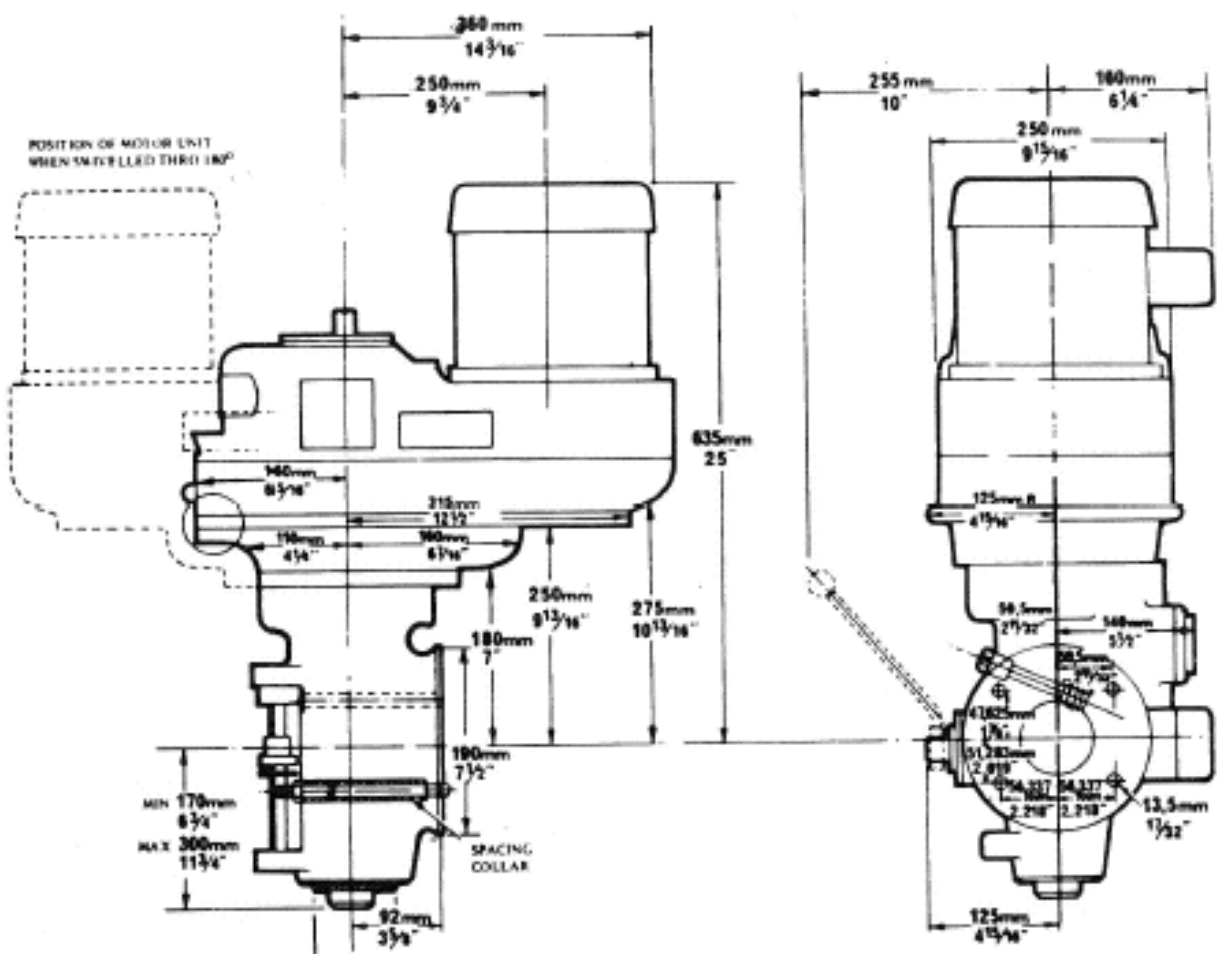
Power feed can be used for drills up to .375"Ø. Use manual feed for drills larger than .375"Ø.

GENERAL SPEED RECOMMENDATIONS

MATERIAL TO BE CUT	FEET PER MINUTE		
	ROUGH CUT	ROUGH AND FINISH	LIGHT AND FINISH CUT
Cast Iron – Soft (Under 150 Brinnell)	70	80 – 90	120
Cast Iron – Med (150-200 Brinnell)	55	60 – 70	90
Cast Iron – Hard (Over 200 Brinnell)	40	50 – 60	70
Steel (Chrome Nickel 40-45 Shore)	30	40	50
Steel (Stainless)	60	80	90
Steel (Low Carbon)	80	90	140
Steel (High Carbon)	40	50	70
Bronze (Medium)	90	120	150
Bronze (Hard)	65	90	130
Brass (Hard)	100	150	200
Copper	150	200	300
Duraluminum	400		600
Aluminum	600		1000

TABLE OF CUTTING SPEEDS AND FEEDS

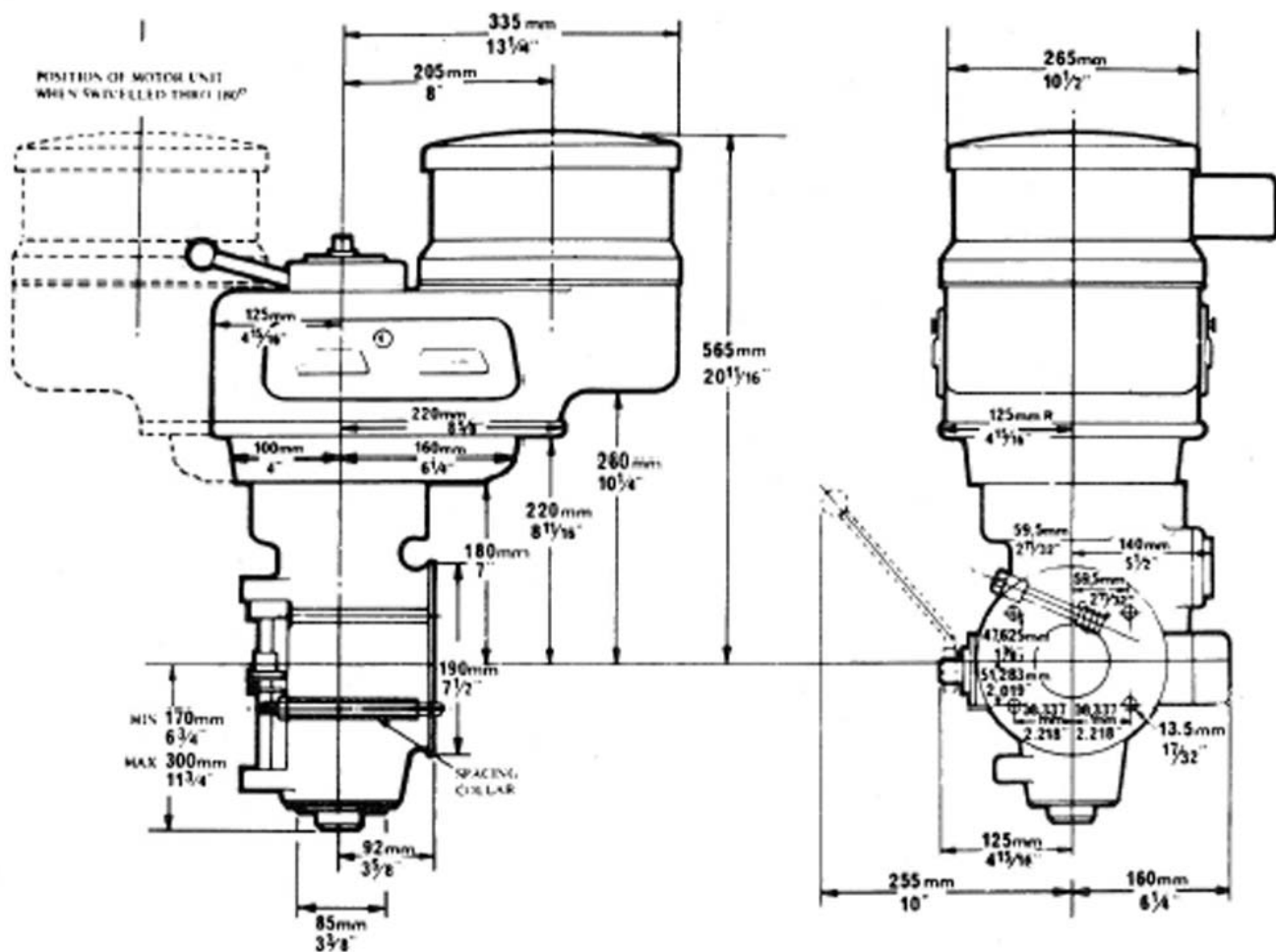
Feet per Minute	15	20	25	30	40	50	60	70	80	90	100
REVOLUTIONS PER MINUTE											
in Inches											
.063"	917	1222	1528	1833	2445	3056	3667	4278			
.125"	458	611	764	917	1222	1528	1833	2139	2445	2750	3056
.187"	306	407	509	611	815	1019	1222	1426	1630	1833	2037
.250"	229	306	382	458	611	764	917	1070	1375	1375	1528
.312"	183	244	306	367	489	611	733	856	978	1100	1222
.375"	153	204	255	306	407	509	611	713	815	917	1019
.438"	131	175	218	262	349	437	524	611	698	786	873
.500"	115	153	191	229	306	382	458	535	611	688	764
.625"	91	122	153	183	244	306	367	428	489	550	611
.750"	76	102	127	153	204	255	306	357	407	458	509
.875"	65	87	109	131	175	218	262	306	349	393	437
1.000"	60	76	95	115	153	191	229	267	306	344	382
1.125"		67	84	102	136	170	204	238	272	306	340
1.250"		61	76	91	122	153	183	214	244	275	306
1.375"			69	83	111	139	167	194	222	250	278
1.500"			63	76	102	127	153	178	204	229	255
1.625"			60	70	94	118	141	165	188	212	235
1.750"				65	87	109	131	153	175	196	218
1.875"				61	81	102	122	143	163	183	204
2.000"					76	95	115	134	153	172	191



TP5268

Figure 5.6 - 2J-Head Specifications

Spindle Taper	R8	R8
Spindle Speeds – RPM	60-4200	60-4200
Motor	*2 HP	1.5 kw
Quill Travel	5"	127mm
Power Feed of Quill	.0015"	.04mm
per Rev of Spindle (3 Rates)	.003"	.08mm
	.006"	.15mm
Collet Capacity	.125" –.750" x .063"	3 – 19mm x 1.5mm
Weight	196 lb.	89 kg
*2 HP Continuous – 3 HP Intermittent		



TP5269

Figure 5.7 - J-Head Specifications

Spindle Taper
Spindle Speeds – RPM

R8
Low Range
80, 135, 210, 325

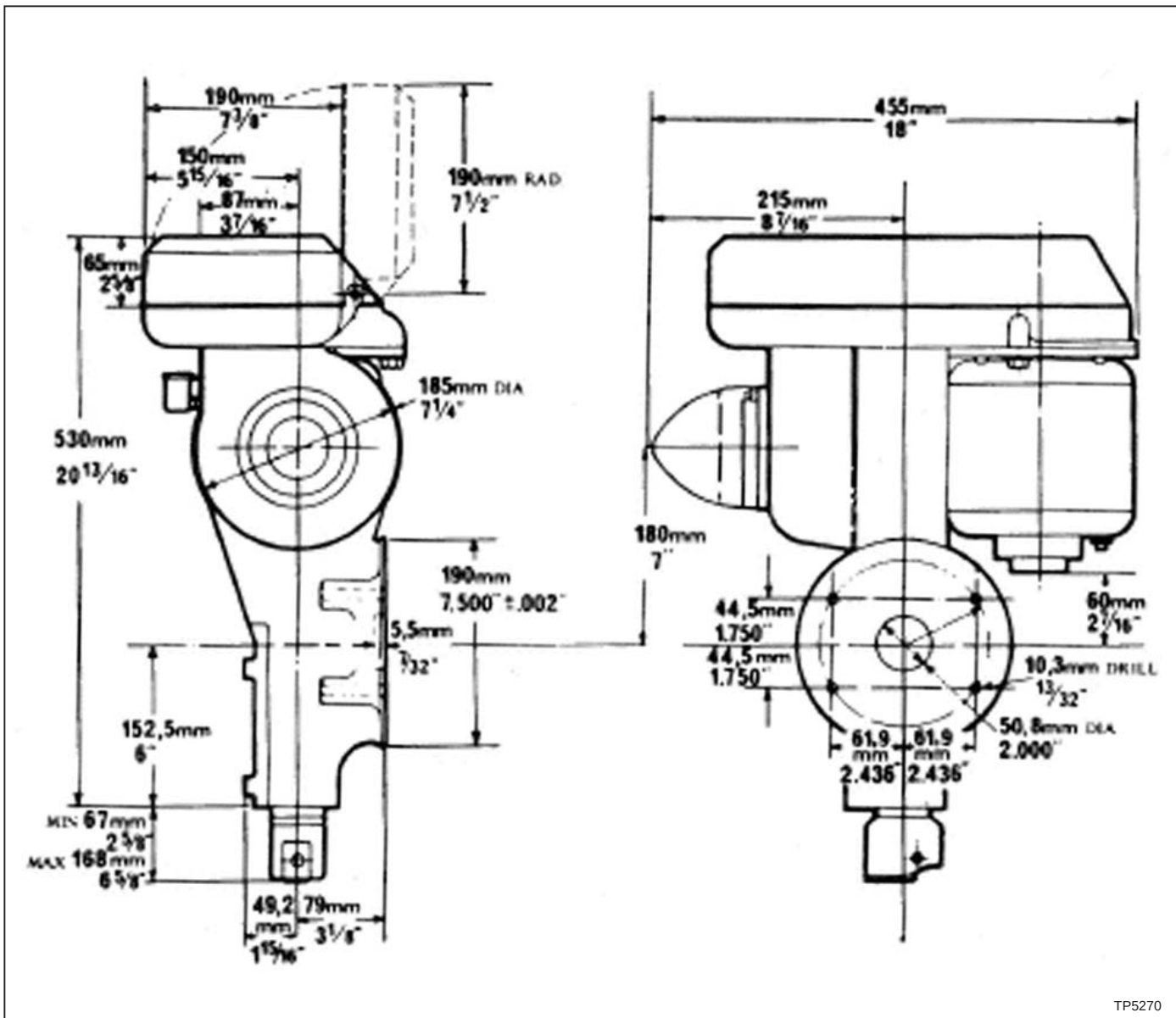
R8
High Range
660, 1115, 1750, 2720

Motor
Quill Travel
Power Feed of Quill
per Rev of Spindle (3 Rates)

1 HP
5"
.0015"
.003"
.006"
.125" – .750"
168 lb

.750 kw
127mm
.05mm
.08mm
.15mm
by .063" increments
76 kg

Collet Capacity
Weight

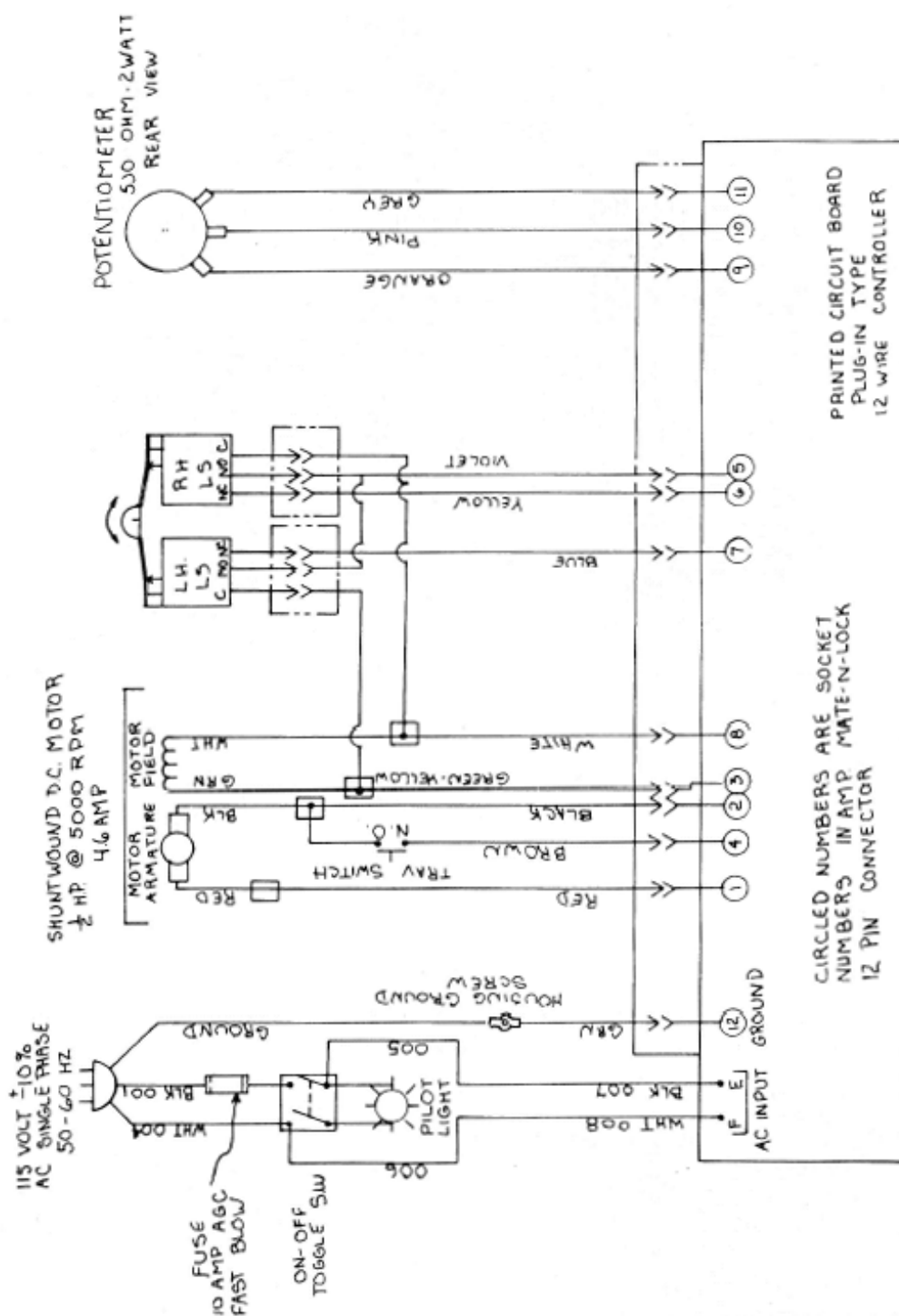


TP5270

Figure 5.8 - E-Head Specifications

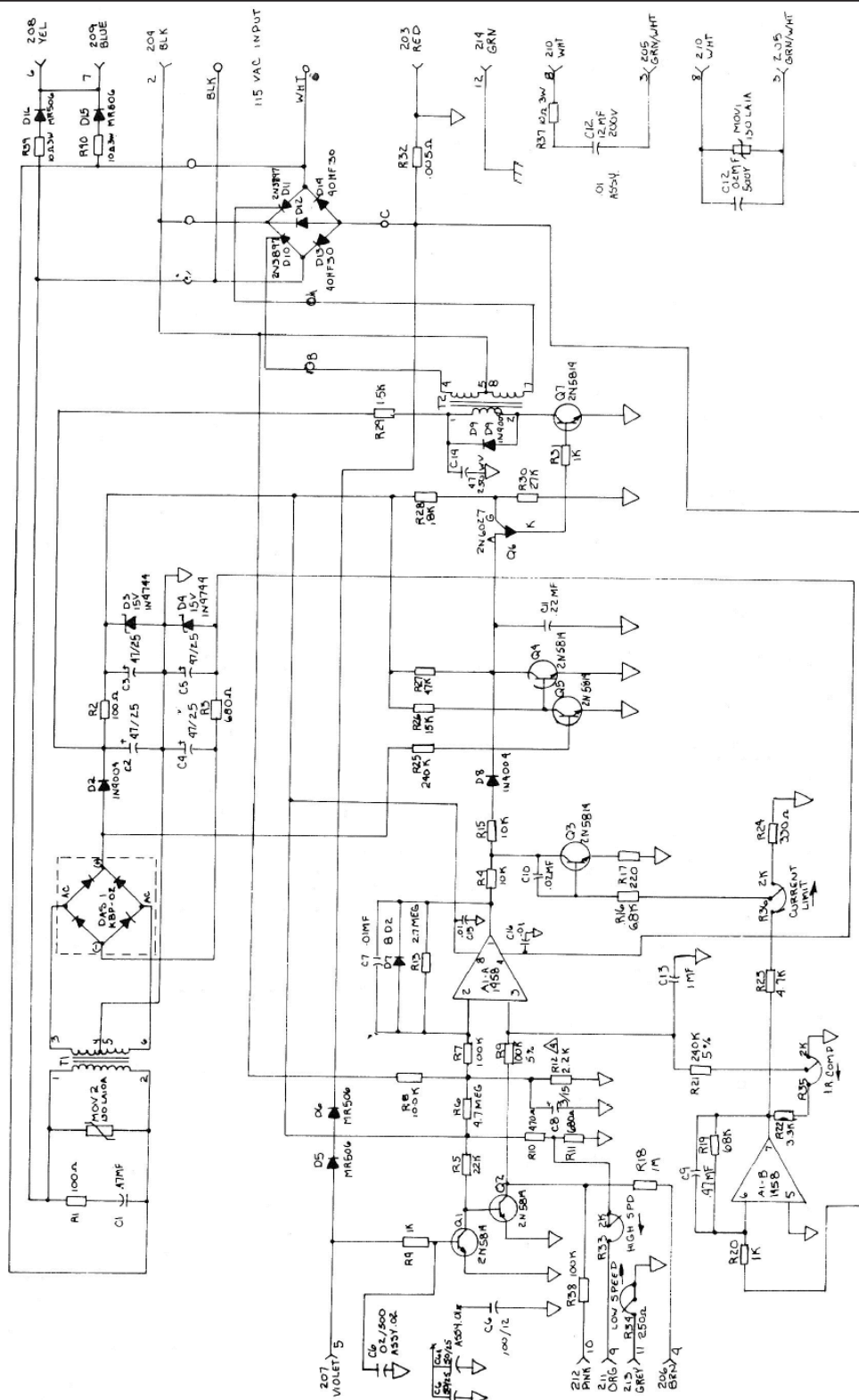
Oscillation Rate (Stroke per Minute)	60, 85, 120 170, 245, 350	
Motor	1/3 HP	.250 kw
Stroke Adjustment	.125" – 4"	3 - 100mm
Tool Capacity		
Horizontally Mounted	.375"	10mm
Vertically Mounted	.625"	16mm
Weight	112 lb	51 kg

SCHEMATIC



TP5271

Figure 5.9 - 6F Power Feed Schematic



TP5272

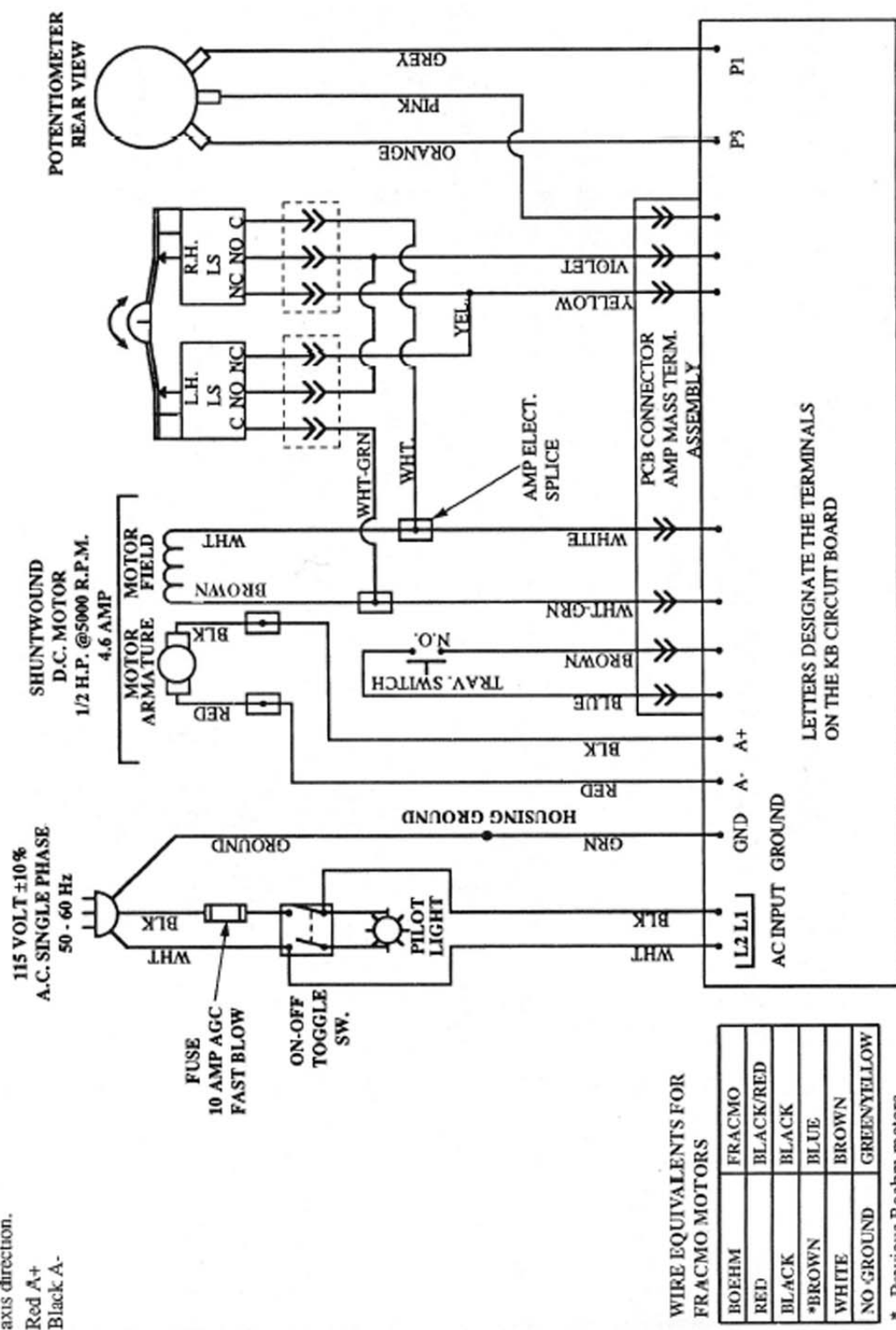
Figure 5.10 - 6F Power Feed Circuit Schematic

SCHEMATIC

NOTE:

For Y-axis power feed, reverse wires for proper axis direction.

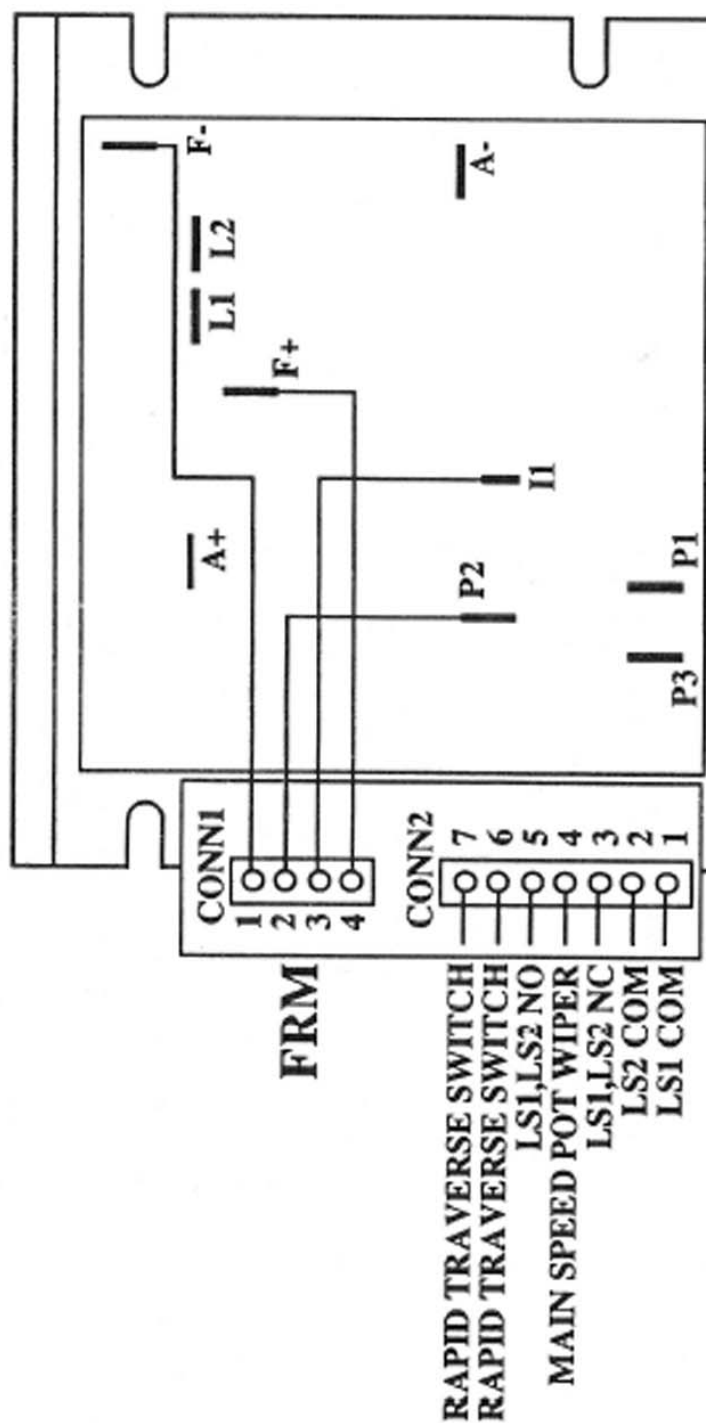
Red A+
Black A-



* Previous Boehm motors used a green field wire.

Figure 5.11 - 8F Power Feed Schematic

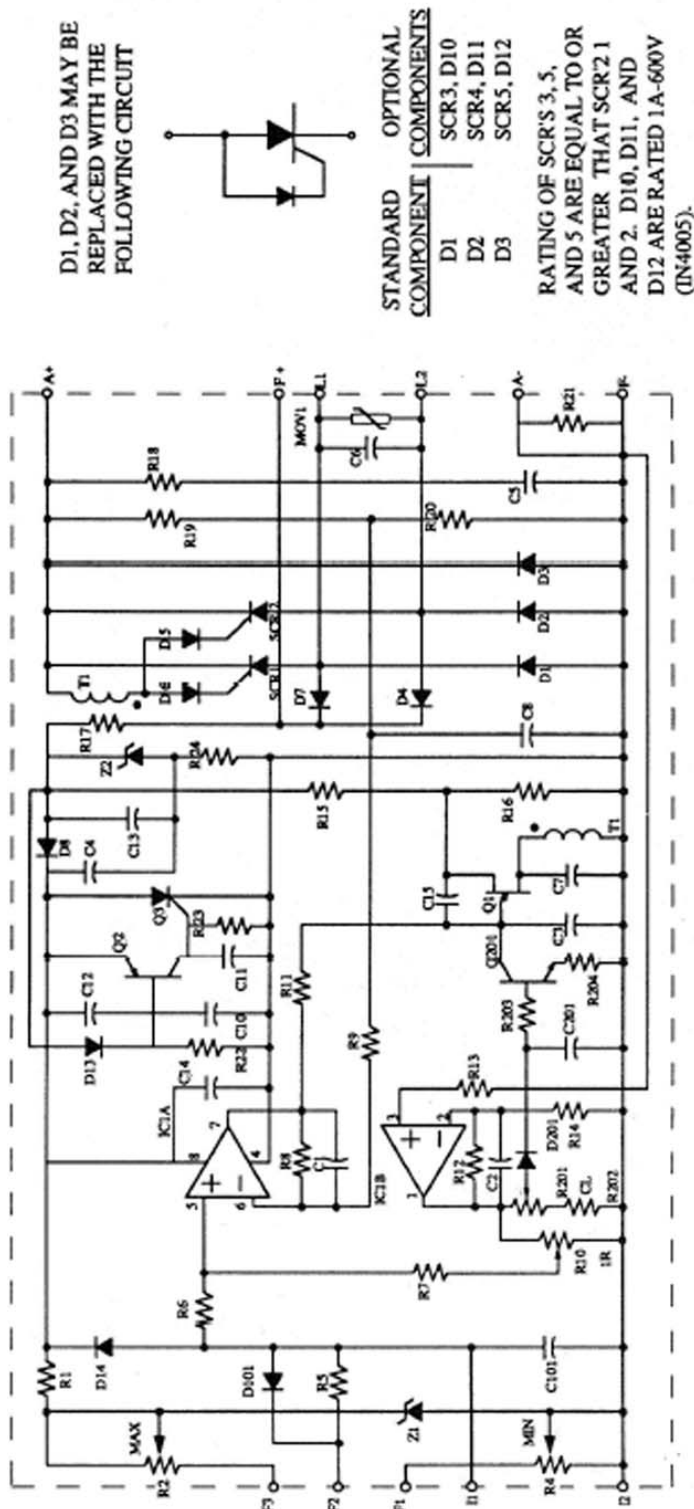
KBIC-118



TP5274

Figure 5.12 - 8F Power Feed Circuit Board Schematic

KBIC-L SCHEMATIC



NOTES:

1. THIS SCHEMATIC APPLIES TO THE FOLLOWING MODELS: KB1C-19, KB1C-125, KB1C-118, KB1C-120, KB1C-29, KB1C-225, KB1C-218, KB1C-240, AND KBPC WITH ONE OR MORE OF THE FOLLOWING SUFFIXES: A, T, U, S, AND PM.
2. R16 MAY BE CHANGED TO Z3 ON SOME MODELS.
3. R3C CAN BE A FIXED RESISTOR.
4. R IS USED ON SOME MODELS.
5. R21 MAY BE A PLUG-IN HORSEPOWER RESISTOR.

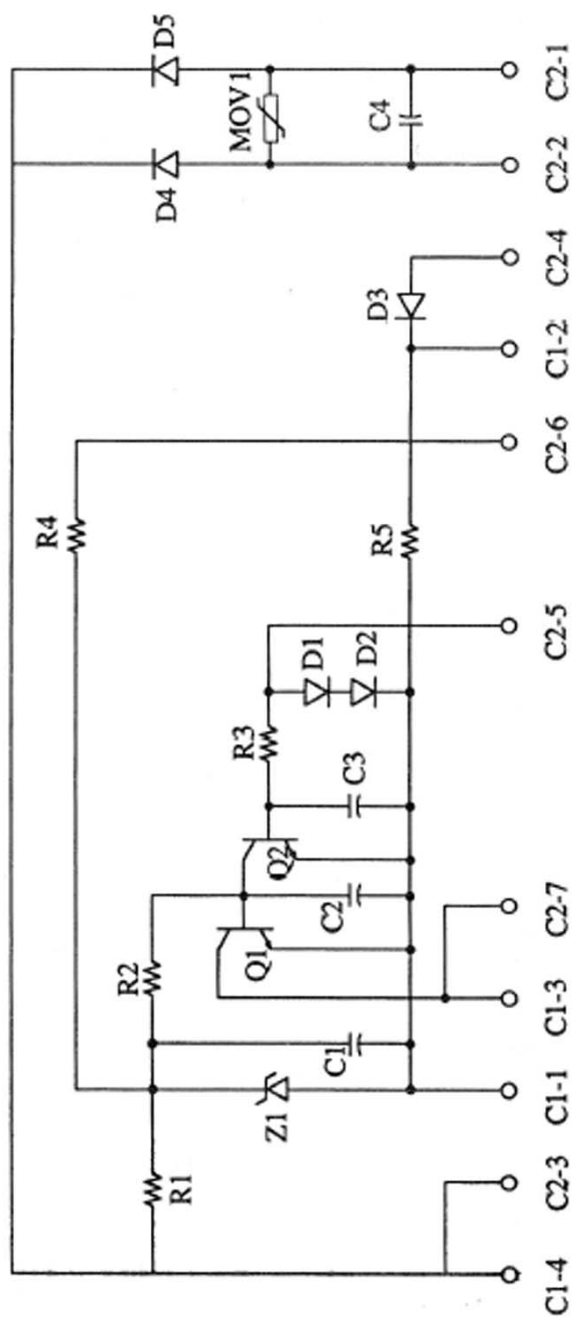
Figure 5.13 - 8F Power Feed Schematic

TP5275

8F POWER FEED SCHEMATIC

Circuit Ref	Value/Rating	Mfr Type	Function
C1	.22ufd – 100V		Capacitor
C2, 3, 8	.1ufd – 100V		Capacitor
C4	100ufd – 35V		Capacitor
C5, 6	.1ufd – 250V		Capacitor
C7	.0033ufd – 100V		Capacitor
C10, 11	4.7ufd – 25V		Capacitor
C12, 13, 15	.01ufd – 25V	Disc Ceramic	Capacitor
C14	.02ufd – 50V	Disc Ceramic	Capacitor
C101	4.2ufd – 25V		Capacitor
C201	.1ufd – 100V		Capacitor
D1, 2, 3	15A – 600V	D6015L	Power Diode
D4 – D14	1A – 600V	1N4005	Diode
D101, 201	1A – 500V	1N4005	Diode
IC1		CA358 or LM358	Dual Operational Amp
MOV1	130V – 10J	V130LA10A	Transient Suppressor
Q1		2N3646	Unijunction transistor
Q2	1A – 35V	2N4403	Transistor (Al)
Q3	.8A – 50V	S051E	Signal SCR
Q201	.5A – 35V	2N5089	Transistor (CL)
R1	2.2K – 1/4W – 10%		Resistor
R2	10K – 1/8E – 20%		Maximum Speed Trim
R3	5K – 2W – 20%		Main Speed Pot
R4	2.5K – 1/8W – 10%		Minimum Speed Trim
R5	100K – 1/4W – 10%		Resistor
R6	10K – 1/4W – 10%		Resistor
R7	150K – 1/4W – 10%		IRC Range
R8*, 203	3.3M – 1/4W – 10%		Resistor
R9	47K – 1/4W – 10%		Resistor
R10	50K – 1/8W – 10%		IR Compensation Trim
R11	33K – 1/4W – 10%		Resistor
R12	2.2M – 1/4W – 10%		Resistor
R13, 14	47K – 1/4W – 10%		Resistor
R15	3.3K – 1/4W – 10%		Resistor
R16	5.6K – 1/4W – 10%		Resistor
R17	4.7K – 3W – 10%		Resistor
R18	470 ohm – 1/4W – 10%	Flameproof	Resistor
R19	68K – 1/4W – 10%		Resistor
R20	3.9K – 1/4W – 10%		Resistor
R21	.01 – 1.0 ohm – 5W – 15%		Current Sensing Resistor
R22	47K – 1/4W – 10%		Resistor
R23	1K – 1/4W – 10%		Resistor
R201	50K – 1/8W – 10%		Current Limit Trim
R202	8.2K – 1/4W – 10%		Resistor
R204	4.7K – 1/4W – 10%		Resistor
SCR1, 2	15A – 200V	S2015L	Power SCR
T1	1:1	KB	Pulse Xfmr
Z1	12V – 1W – 5%		Zener Diode
Z2	22V – 1W – 5%		Zener Diode

FRM, SCHEMATIC DIAGRAM



TP5276

Figure 5.14 - 8F Power Feed Circuit Board Schematic

8F POWER FEED CIRCUIT BOARD

Circuit Ref	Value/Rating	Mfr Type	Function
C1, 3	10.0ufd - 35V	Electrolytic	Capacitor
C2	0.47ufd - 50V	Electrolytic	Capacitor
C4	0.022ufd - 250Vac	Mylar CE156F20-4	Capacitor
Conn. 1	/630Vdc		End Connector
D1, 2, 3	1A - 600V	1N4005GP	Diode
D4, 5	1.5A - 600V	1N5937GP	Diode
HDR1		MLSS156-4	PCB Header
HDR2		MLSS156-7	PCB Header
MOV1	150V	V150LA10AA	Transient Surge Supp
Q1, 3	0.6A - 1W - 5%	2N4401	Small Signal Transistor
R1	33K - 1W - 5%	MO-1	Resistor
R2	22K - 0.25W - 5%	Carbon Film	Resistor
R3, 4	10K - 0.25W - 5%	Carbon Film	Resistor
R5	33K - 0.25W - 5%	Carbon Film	Resistor

